

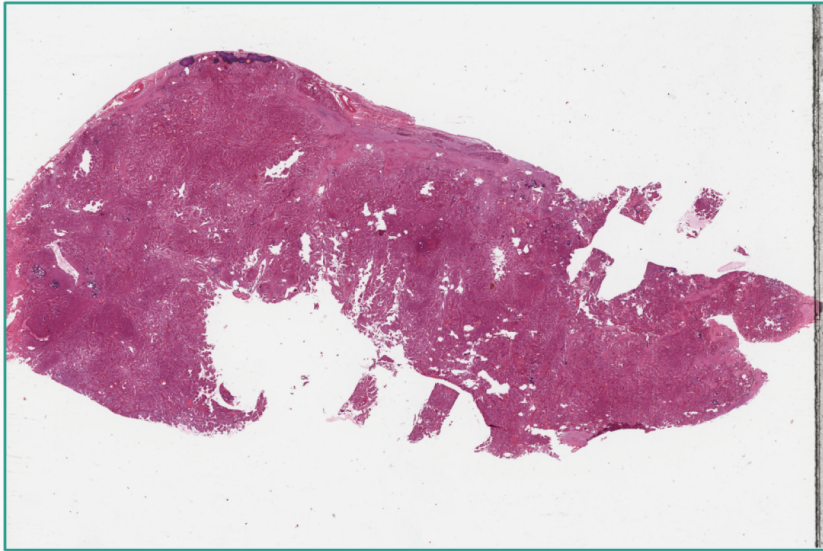
PHASE 1 – Spatial UNI -> DM1 correlation

GSE250521 (n=16 thyroid spatial–transcriptomic slides)

Slides with rho > 0.3	8 / 16
Strongest negative rho	–0.61 (LPTC–2)
Significant slides (p < 0.05)	11 / 16
PC1 explained var (mean)	27.4%

Spatial signal: DM1 score tracks UNI tile embeddings within a slide.

BEST DM1+ ATTENTION (CLAM INFERNO heatmap)



slide c3753183... | DM1 prob = 0.97 | fold 1

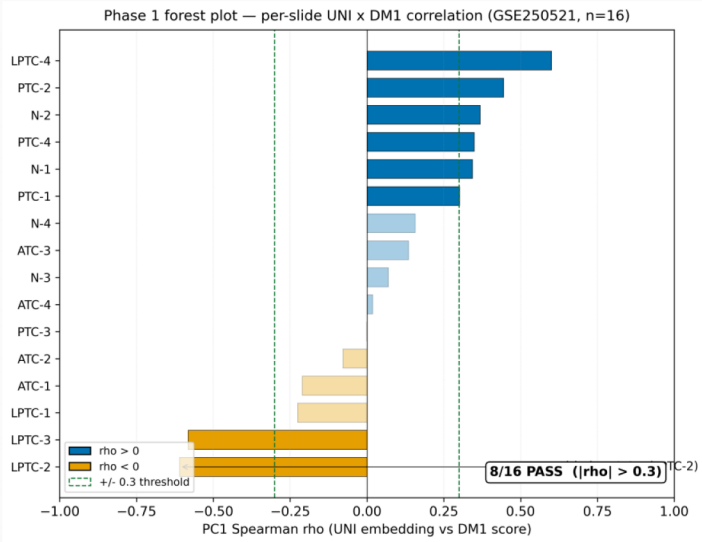
PHASE 2 – CLAM held–out classifier

TCGA–THCA (n=88 slides, 5–fold cross–validation)

Pooled cross–fold AUC	0.746
95% CI	[0.61 – 0.86]
Mean per–fold AUC	0.830 +/- 0.139
All folds AUC	>= 0.71 (2/5 = 1.00)
ResNet50 baseline	~0.55 (negative)

All folds clear PASS threshold (0.70). UNI features beat ImageNet baseline.

PHASE 1 PER–SLIDE rho FOREST (subgroups)



Per–slide rho: –0.61 to +0.34; Stage I subgroup AUC trends similar to overall.